

Léo Mans



 Guangzhou, China

PROFILE

A soon-to-graduate Master's student passionate about emerging technologies, particularly Mixed Reality. Throughout my studies, I have developed and contributed to multiple research projects focusing on Virtual Reality and HCI. I am currently seeking a Ph.D. position specializing in human-centered immersive technologies for Spring 2027. I aim to leverage my technical background to create intuitive experiences that elevate human potential.

EDUCATION

Master's Degree in Industrial Engineering

2024-2026

Arts et Métiers, Chalon-sur-Saône, France

- **Specialization:** Management of Interactive 3D Technologies
- **Core coursework:** Virtual Reality, Augmented Reality, Image Analysis, 3D Modeling, BIM, Lean Engineering, Research Methodologies, Industry 4.0
- **Master Thesis:** "Bacteria Visualization in a Kitchen Environment" Conducted at HKUST-GZ under the supervision of Prof. Wang and Prof. Chardonnet
- **Overall Grade:** 15/20

Bachelor's Degree in Computer Science

2021-2024

University of Lorraine, Vandoeuvre-lès-Nancy, France

- **Core coursework:** Robotics, Cybersecurity, Web & Software Development, Networks, Database Management

RESEARCH INTERESTS

- **Affective Computing:** Understanding, recognizing, and enhancing human emotions and affective states through AI and XR technologies.
- **Social Virtual Reality (Social VR):** Investigating human behavior, social interactions, collaboration, communication, and group dynamics in immersive virtual environments.
- **Human Augmentation and Accessibility:** Leveraging AI and XR to enhance human performance, improve cognitive and physical capabilities, and increase accessibility and inclusion for diverse populations.
- **Human-Computer Interaction (HCI):** Designing, developing, and evaluating intuitive, immersive, and user-centered interactive systems that improve user experience and usability.

RESEARCH EXPERIENCE

Visiting Research Student

Feb 2026-Sep 2026

The Hong Kong University of Science and Technology

Guangzhou, China *Advised by Prof. Wang and Prof. Chardonnet*

- **Research Objective:** Investigated novel HCI paradigms for visualizing invisible pathogens in immersive environments to evaluate the impact of cognitive load and temporal pressure on hygiene compliance within professional kitchens.
- **Technical Development:** Engineered an interactive VR serious game simulating a commercial kitchen "rush," implementing three distinct pathogen visualization modalities (animated particles, static texture modification, and flashing alerts) alongside environmental stressors.
- **Methodology & User Testing:** Designed a progressive stress-based experimental protocol. Evaluated the efficacy of the visual interfaces by correlating objective performance telemetry (cross-contamination errors) with physiological stress data (heart rate monitors) and perceived cognitive load (NASA-TLX).
- **Expected Outcomes:** Formulating ergonomic design guidelines for representing invisible biological hazards in XR, aiming to find the optimal balance between visual awareness and cognitive limits during high-stress operational training.

SELECTED ACADEMIC PROJECTS

Cybersickness Study

Jan 2026

Arts et Métiers, Chalon-sur-Saône, France *Advised by Prof. Merienne*

- Established a scientific protocol (state-of-the-art review, VR app development, experimentation, and data analysis) and authored a 2-page scientific paper within a 4-day deadline.

TerrariskVR: Rockfall Risk Awareness

Jan 2026

GRDF, Troyes, France

- Developed an educational VR application for rockfall risk awareness and produced an accompanying promotional video.

Guit'AR: An Augmented Reality Tool for Learning Guitar Scales

Nov 2025

Conservatoire du Grand Chalon, Chalon-sur-Saône, France *Advised by Prof. Chardonnet and Prof. Ackerman*

- Engineered an educational Augmented Reality application that displays a virtual 3D guitar to visualize interactive grid scales and optimal finger positioning.
- Conducted a state-of-the-art on AR applications in music education and led an experimental study with guitar students at the Conservatoire.

Franco-German Hackathon

Oct 2025

Arts et Métiers, Paris, France *Organized by Prof. Chardonnet and Prof. Langbehn*

- Developed a VR application within an international team to raise awareness about ocular conditions (astigmatism and color blindness).

LAMIS: A VR Escape Room for Mathematics Learning

Apr 2025-Jul 2025

University of Clermont Auvergne, Aubière, France *Advised by Prof. Chardonnet and Prof. Cogneras*

- Engineered a cross-platform educational Escape Room focused on mathematics using WebXR, ensuring accessibility across both fully immersive VR headsets and standard web browsers.
- Conducted a state-of-the-art review on academic learning in virtual environments. Evaluated the application's efficacy using System Usability Scale (SUS) questionnaires and qualitative user feedback to assess perceived educational progress and overall usability.

PVD VR: Understanding the PVD Process Through Virtual Reality

Mar 2025

Femto-ST Institut, Besançon, France *Advised by Prof. Lou and Prof. Besnard*

- Implemented a 3D model based on a 3D scan, remeshed using MeshLab, optimized in 3ds Max, and modified meshes for animation.
- Conducted a state-of-the-art review on applications visualizing invisible processes, and designed an experiment to compare learning paradigms (PDF course, video, VR application).

VrTech: Discovering a Glass Bottle Production Line

Jan 2025

Vertech', Chalon-sur-Saône, France

- Developed a VR training and discovery application simulating a glass manufacturing production line.

PROFESSIONAL EXPERIENCE

Virtual Reality Engineering Intern

Apr 2025-Jul 2025

University of Clermont Auvergne, Aubière, France

- Designed a virtual reality application aimed at facilitating the learning of mathematical concepts through a playful and immersive experience.

Software Development Intern

Mar 2024-Jun 2024

Jacques PREVOT Artifices, Sarrey, France

- Designed a decision-support algorithm to automatically recommend the optimal carrier for shipping orders containing hazardous materials, considering logistical and regulatory constraints.

TECHNICAL SKILLS

Virtual Reality & Augmented Reality: OpenXR, MetaSDK, WebXR, Vuforia

3D Modeling & Rendering: 3ds Max, Blender, SketchUp, SolidWorks, MeshLab, Unreal Engine 5, Revit

Game & Software Development: Unity, Android Studio, SFML, SDL2

Image Analysis: Python, OpenGL, MediaPipe, YOLO

Hardware: 3D Printing, Arduino, ESP32

Photo & Video: Photoshop, Photopea, Premiere Pro, DaVinci Resolve

Programming Languages: C#, C, C++, Python, Java

LANGUAGES

French: Native

English: B1

Spanish : Notions